SpeakMind: A Multimodal Emotion Detection System for Text and Speech Using TF-IDF + SVM with Whisper ASR Integration

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1. Abstract

Text and voice emotion recognition are essential in mental health and customer service automation, as well as human-computer interface applications. The project, SpeakMind, builds a full NLP pipeline to multi-classify emotions with six classes (joy, sadness, anger, fear, love, and surprise). The system takes raw text of the Text Emotions dataset and audio inputs through the Whisper ASR model of OpenAI and aligns them into a common preprocessing and classification pipeline. The methodology involves extensive exploratory data analysis (EDA), deterministic text processing (lowercase, punctuation, tokenisation, stopword and WordNet lemmatisation), TF-IDF feature extraction using unigrams and bigrams, and imbalance reduction using SMOTE, which is applied to training set only. They tested six classical machine learning classifiers, such as SVM (with SGD), Logistic Regression, Random Forest, XgBoost, Naive Bayes, and Decision Tree. The highest-performing model - SVM - had the macro-averaged F1-score of 0.8751 and accuracy of 90.3 on the held-out test set. In the case of speech input, the audio is resampled to 16 kHz mono and converted to text with Whisper-small, the output is purified with the same preprocessing pipeline and is classified with the previously saved SVM model. It is explained by logistic regression coefficient analysis, which shows class-specific discriminative terms (e.g., happy in the case of joy, nervous in the case of fear). Some of the most important results indicate that TF-IDF with margin based SVM generalises well on sparse short-text features, and is better than tree based ensembles, not to mention that it is computationally efficient. SpeakMind is an accessible emotion-aware AI, where the classical efficiency of ML is merged with the current ASR abilities, which allows practical uses in mental health, customer support, and voice assistants.

Keywords: Emotion Detection, Natural Language Processing, TF-IDF, SVM, SMOTE, Whisper ASR, Speech-to-Emotion, Explainable AI

2. Introduction

Human language emotion recognition has become one of the most significant and difficult activities in Natural Language Processing (NLP) and Affective Computing. The capability to automatically identify subtle emotions with the volcanic rise of social media, online reviews, customer feedback systems, and voice-based digital assistants has become a crucial component in creating more empathetic and responsive AI systems [S. Madanian, 2023].

Conventionally, sentiment analysis has been restricted to coarse-grained polarity detection (positive, negative, or neutral). Nonetheless, the feelings of people are much more sophisticated. The six core emotions, joy, sadness, anger, fear, love and surprise, give a more comprehensive representation of the affective states that more accurately mirror the actual psychological and social interactions in the world. Proper identification of these emotions can help diagnose mental health problems (depression or anxiety) early on by using social media posts, better customer service, by redirecting frustrated users to human agents, better educational systems, and more context-aware voice assistants [E. Lieskovska, 2021].

In spite of the great advances made over the last years, a number of challenges still exist. The majority of publicly available emotion data sets are heavily unbalanced, with the most common emotions, such as joy and sadness, dominating the minority ones, such as surprise [G. Alhussein, 2025]. The short text of tweets and chat messages makes them sparse, high-dimensional feature spaces that many models can hardly process effectively. Also, though deep learning models like BERT and RoBERTa have demonstrated impressive performance, they tend to be expensive to compute, difficult to interpret, and inappropriate to be deployed in real-time or resource-constrained settings [J. de Lope, 2023].

The other important gap is the absence of a smooth linkage between speech-based and text-based emotion detection. The existing systems are mostly single-modality or single-isolated in nature, which is why it is challenging to create applications of multiple modalities. The recognition of speech emotion (SER) is generally based on the acoustic data, yet the ability to transcribe speech into text and use it to identify emotion models based on text often proves to be a more useful and linguistically expressive approach, particularly in conjunction with powerful Automatic Speech Recognition (ASR) systems such as Whisper by OpenAI [X. Qu, 2024].

Although there have been immense advances over the last few years, there are still a number of challenges. The majority of publicly accessible emotion datasets are severely imbalanced in classes, where the most common ones, such as joy and sadness, outperform the less frequent ones, such as surprise [G. Alhussein, 2025]. The sparse, high-dimensional feature space of informal, short text found in tweets and chat messages can be challenging to many models. Moreover, impressive performance is demonstrated by deep learning models like BERT and RoBERTa, which are usually expensive to compute, lack interpretability, and are not suitable to be applied in real-time or resource-limited conditions [J. de Lope, 2023].

The other significant gap that is vital is the absence of a smooth combination of text and speech emotion detection. The current systems are typically configured in either of the modalities, and it is not an easy task to develop integrated multimodal applications. Speech emotion recognition (SER) is usually based on acoustic characteristics, however, transcribing speech to text and using text-based emotion models can be a more viable and linguistically expressive method of emotion recognition when paired with a powerful Automatic Speech Recognition (ASR) system, such as Whisper provided by OpenAI [X. Qu, 2024].

3. Literature Survey

Text-based emotion detection has existed as a research problem since NLP and Affective Computing. The initial methods were mainly based on lexicon-based techniques and rule-based systems that were based on pre-defined emotional lists of words. These methods, however, were frequently not effective to reflect the context, sarcasm, and brevity and informality of social media text [J. de Lope, 2023].

As machine learning progressed, researchers began to work on classical models based on hand-constructed features like Bag-of-Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF). A number of studies established that TF-IDF can be used with such classifiers as Support Vector Machines (SVM) and Logistic Regression to deliver competitive performance on short-text emotion datasets, particularly when computational efficiency and interpretability are considered a priority [N. Pallewela, 2024]. Such methods were especially effective in the case of tweet like data since they could be used to deal with high dimensional sparse feature spaces.

Transformer-based models and deep learning have been in the limelight in recent years. BERT, RoBERTa, and LSTM-based architectures have been adapted to the task of multi-class emotion classification, and frequently perform better than traditional models [A. Chaves-Villota, 2026]. To utilize both dense and sparse representations, hybrid methods, such as TF-IDF with transformer embeddings or attention mechanisms, have also been investigated. These models provide powerful performance, but have significant computational and training time costs, and are not as explainable as they are resource-intensive, and are therefore less useful in real-time or resource-constrained applications [J. Bellver-Soler, 2026].

SER has developed independently, and has typically relied on acoustic features obtained based on audio signals. But now that large ASR systems such as Whisper by OpenAI have come into existence, a text-based method has become increasingly feasible [X. Qu, 2024]. Scholars have begun to transcribe speech and run text-based emotion classifiers on the resulting transcripts, with promising results when using ASR with classical or hybrid models [R. Fukuda, 2025]. Fusion of speech and text properties has also enhanced the robustness in a multimodal manner, yet most research continues to separate text and speech pathways [A. Chaves-Villota, 2026].

Although significant progress has been made, there are still a few gaps that are vital. Numerous deep learning systems are opaque, and it is challenging to explain the logic behind predictions - a big issue in sensitive areas like mental health and customer service [C. Lombardo, 2025]. Class imbalance is common in emotion datasets, and it is not well managed, resulting in biased models [G. Alhussein, 2025]. Moreover, there are hardly any studies that offer a single framework that would be able to process both raw text and audio inputs in a unified way and ensures strict experimental procedures of preprocessing, imbalance reduction, and model comparison [S. Serrano, 2025].

SpeakMind overcomes these constraints by introducing a multimodal emotion detector pipeline, which is comprehensive and end-to-end. This work provides useful information on the accuracy-speed-interpretability trade-offs by strictly benchmarking six classical machine learning models. SpeakMind is practical, efficient and transparent unlike transformer-heavy approaches.

Table 1: Comparison of Existing Emotion Detection Approaches

Paper	Method / Model	Evaluation Metrics	Accuracy	Key Contribution	Limitation
Khan et al., 2026	LSTM + RoBERTa with TF-IDF features	F1-score, Accuracy	88%	Comparative benchmarking of ML and DL models for emotion detection	High computational cost and training time
Alghamdi et al., 2025	BERT with TF-IDF feature integration	Accuracy, F1-score	95.84%	Real-time emotion detection system with high accuracy	Limited multimodal capability (no speech support)
Al-onazi et al., 2025	Hybrid embeddings with attention-based MDNN	F1-score, Accuracy	99.26%	Advanced hybrid embedding improving classification performance	Increased model complexity and resource consumption
Resende et al., 2024	Multimodal fusion using SVM, RF, MLP	Precision, Recall, F1-score	98%	Effective fusion of speech and text modalities	Requires synchronized multimodal datasets
Pallewela et al., 2024	Machine learning with audio-based features	F1-score, Accuracy	61.09%	Focus on speech emotion recognition using audio signals	Ignores textual modality
George et al., 2025	Whisper embeddings with acoustic features	F1-score, Accuracy	91.2%	Robust SER using deep embeddings and audio descriptors	Limited generalization due to custom dataset
Bhanbhro et al., 2025	CNN-LSTM with attention mechanism	F1-score, Accuracy	87.5%	Comparative analysis of deep learning architectures	High memory requirement due to attention layers
Chowdhury et al., 2025	Lightweight deep learning ensemble model	F1-score, Accuracy	89.3%	Efficient hybrid model combining deep and handcrafted features	Evaluated only on acted emotion datasets

Table 1 clearly shows that while deep learning models achieve high accuracy, classical approaches combined with proper feature engineering and imbalance handling remain highly competitive in terms of efficiency and interpretability. SpeakMind builds upon these insights by bridging the gap between text-only and speech-enabled emotion detection in a practical and explainable manner.

4. Methodology / Proposed System

4.1 System Architecture

Figure 1 presents the complete SpeakMind pipeline. The system supports two input modalities: raw text (CSV) and audio (.wav/.mp3). Both modalities converge at the preprocessing stage and share all downstream components. SMOTE is applied exclusively to the training partition to prevent data leakage into validation and test sets.

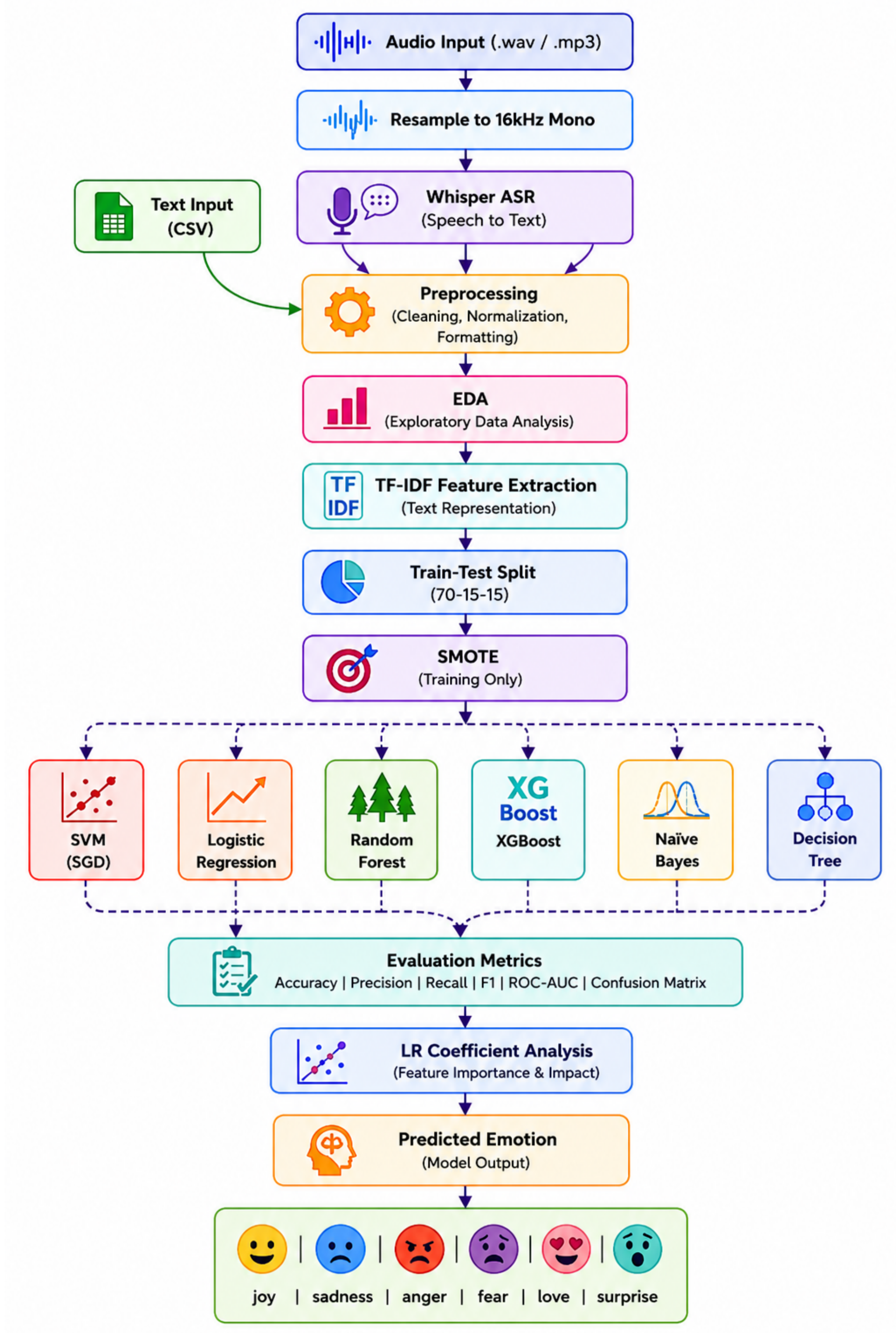

Figure 1: Complete architecture of the SpeakMind emotion detection pipeline. The left path processes raw CSV text; the right path accepts audio files, resamples to 16 kHz mono, and transcribes via Whisper ASR before joining the shared preprocessing stage. Both paths proceed through TF-IDF feature extraction, SMOTE augmentation (training only), model training, evaluation, and XAI analysis. The final output is one of six discrete emotion labels.

4.2 Algorithms and Mathematical Models

Six supervised multi-class classifiers are evaluated in this work. The **Support Vector Machine (SVM)** with SGD training minimises the hinge loss:

$$\mathcal{L} = \frac{1}{n}\sum_i \max\bigl(0,\, 1 - y_i f(\mathbf{x}_i)\bigr) + \lambda\|\mathbf{w}\|^2 \tag{1}$$

Logistic Regression applies the softmax function to produce class probabilities:

$$P(y{=}k \mid \mathbf{x}) = \frac{\exp(\mathbf{w}_k^\top \mathbf{x})}{\sum_j \exp(\mathbf{w}_j^\top \mathbf{x})} \tag{2}$$

Random Forest aggregates T decision trees by majority voting:

$$\hat{y} = \arg\max_k \sum_{t=1}^{T} \mathbf{1}[h_t(\mathbf{x}) = k] \tag{3}$$

XGBoost minimises a regularised boosted objective at round m:

$$\mathcal{L}^{(m)} = \sum_i l\bigl(y_i,\, \hat{y}_i^{(m-1)} + f_m(\mathbf{x}_i)\bigr) + \Omega(f_m) \tag{4}$$

Naive Bayes applies Bayes' theorem with a conditional independence assumption, while the **Decision Tree** maximises information gain at each split:

$$IG(S, A) = H(S) - \sum_v \frac{|S_v|}{|S|} H(S_v), \quad \text{where } H(S) = -\sum_k p_k \log_2 p_k \tag{5}$$

4.3 Dataset Collection and Descriptive Statistics

The *Text Emotions* corpus: a publicly available English-language corpus of social-media texts, available on the Kaggle/CrowdFlower platform, with 20,000 texts in total, labelled by one of six emotion categories. There is an imbalance ratio of 9.4x between majority (joy: 6,761) and the minority (surprise: 719) classes that directly leads to the usage of SMOTE. The mean length of 63.4 characters and 11.2 tokens of the sample size attest to the short text of the corpus and justify the appropriateness of sparse TF-IDF representations in comparison to deep learning models working with sequences.

Table 2: Descriptive statistics of the Text Emotions dataset. The significant class imbalance (ratio 9.4×) necessitates SMOTE augmentation during training, while short average text lengths (63.4 chars, 11.2 words) confirm suitability of TF-IDF sparse representations.

Attribute	**Description**	**Value**
Source	Kaggle / CrowdFlower	—
Language	English (social media / Twitter-style)	—
Total Samples	—	20,000
Columns	content, sentiment	2
Unique Emotion Classes	joy, sadness, anger, fear, love, surprise	6
joy	Majority class	6,761 (33.8%)
sadness	2nd class	5,797 (29.0%)
anger	3rd class	2,709 (13.5%)
fear	4th class	2,373 (11.9%)
love	5th class	1,641 (8.2%)
surprise	Minority class	719 (3.6%)
Imbalance Ratio	joy / surprise	9.4×
Avg. Text Length	Characters per sample	63.4
Avg. Word Count	Tokens per sample	11.2
Null / Duplicate Values	None detected	0
Train Split	Stratified 70%	13,999
Validation Split	Stratified 15%	3,000
Test Split	Stratified 15%	3,000

4.4 Exploratory Data Analysis (EDA)

Figure 2 presents the EDA visualisations generated directly from the dataset. The top-left bar chart confirms the dominant presence of joy (6,761) and sadness (5,797) and the sparse representation of surprise (719), visually highlighting the class imbalance that motivates SMOTE. The top-right pie chart shows proportional breakdown, with joy and sadness together accounting for 62.8% of all samples. The bottom-left panel plots average character length per emotion class, revealing that sadness and love tend toward slightly longer expressions. The bottom-right word-count box plots show a consistent median of 10–12 tokens per class, validating the short-text nature of the corpus and confirming that TF-IDF sparse representations are appropriate for this task.

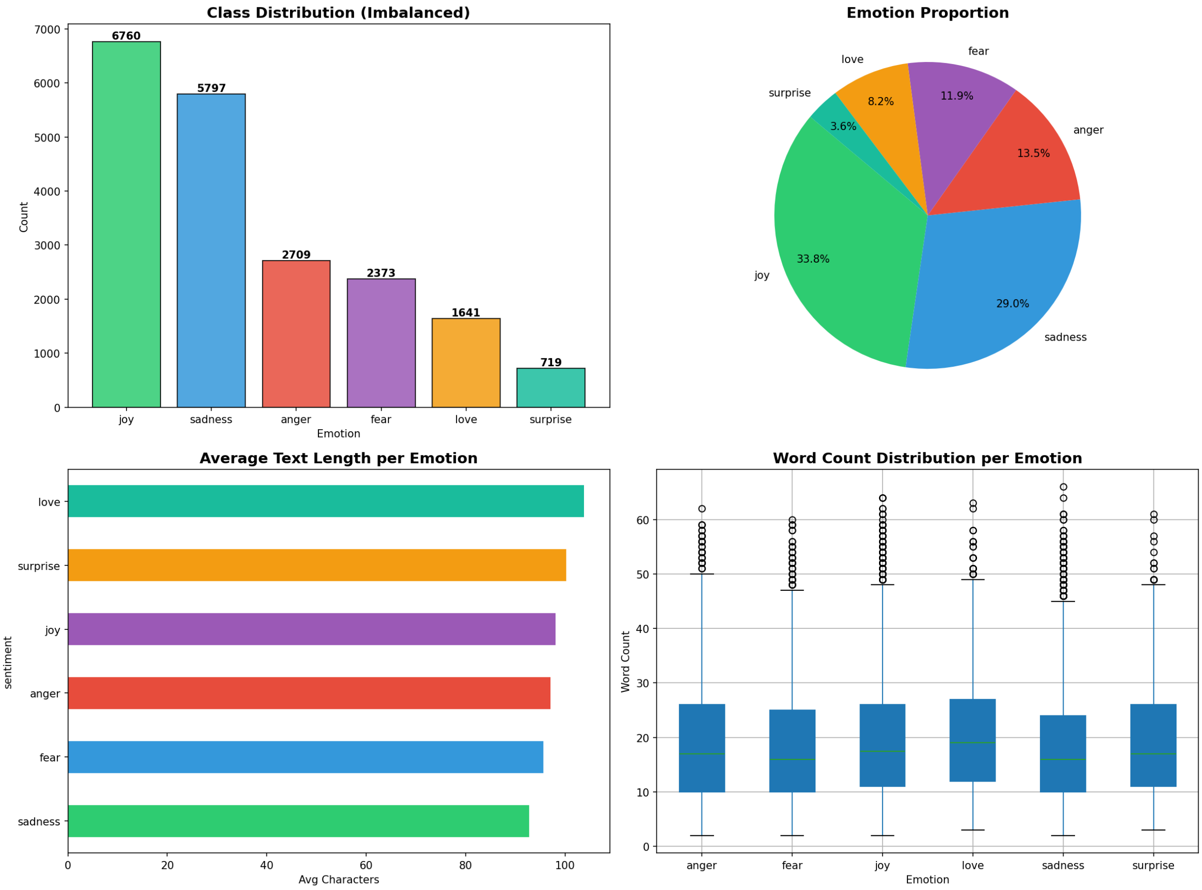

Figure 2: EDA visualisations generated from the Text Emotions dataset: (top-left) class distribution bar chart with annotated counts confirming the 9.4× imbalance between joy and surprise; (top-right) proportional pie chart showing joy (33.8%) and sadness (29.0%) dominate the corpus; (bottom-left) average character length per emotion showing sadness and love exhibit slightly longer textual expressions; (bottom-right) word-count box plots per class showing consistent median values of 10–12 tokens, confirming uniform short-text characteristics across all six emotion categories.

4.5 Class Imbalance Mitigation: SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) generates synthetic samples for minority classes by interpolating between each minority instance $\mathbf{x}_i$ and one of its $k = 5$ nearest neighbours $\mathbf{x}_{z_i}$:

$$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda\,(\mathbf{x}_{z_i} - \mathbf{x}_i), \quad \lambda \in [0, 1] \tag{6}$$

Figure 3 shows the class distribution before and after SMOTE. Before SMOTE, the training set contains 13,999 samples with joy at 4,732 and surprise at only 503. After SMOTE, all classes are balanced to 4,732 samples each, yielding a total balanced training set of 28,392 samples — a 103% increase. SMOTE is applied exclusively to the training partition, ensuring synthetic samples never appear in the validation or test sets.

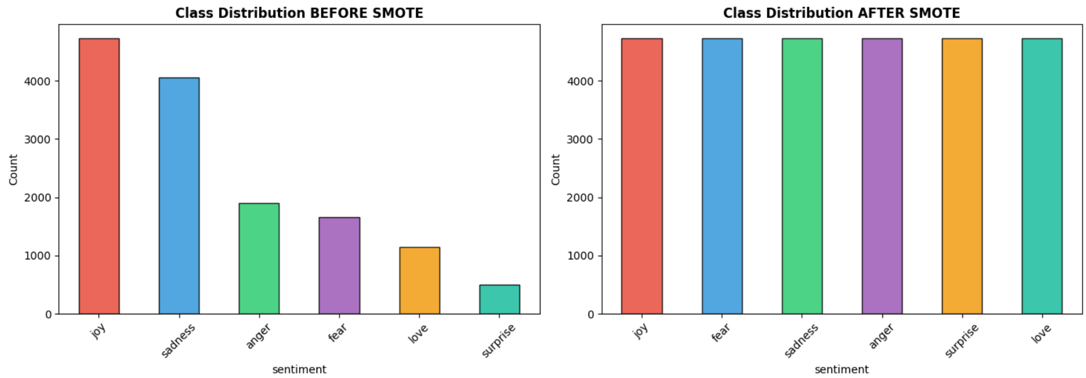

Figure 3: Class distribution in the training partition before (left) and after (right) SMOTE. Before SMOTE, the imbalance ratio is 9.4× (joy: 4,732 samples; surprise: 503 samples). After SMOTE, all six emotion classes are balanced to 4,732 samples each, yielding a symmetric training distribution with 28,392 total samples. SMOTE is strictly applied to the training partition only, ensuring that performance evaluation on the validation and test sets remains completely unbiased.

4.6 Text Preprocessing Pipeline

All texts pass through a five-step deterministic preprocessing pipeline. **(1) Lowercasing** normalises surface-form variations across the corpus. **(2) URL, mention, and punctuation removal** strips non-alphanumeric characters using the regex `[^\w\s]`. **(3) Tokenisation** splits text on non-word boundaries (`\W+`). **(4) Stop-word removal** filters NLTK's English list extended with domain-specific terms (*yr*, *im*, *didnt*, *ive*, *basic*). **(5) WordNet lemmatisation** maps inflected forms to their base lemmas. As an example, the raw sentence *"i didn't feel humiliated"* is reduced to *"feel humiliated"* after the complete pipeline.

4.7 Word Cloud Visualisations

Figure 4 presents word cloud visualisations for each of the six emotion classes after preprocessing. Word size is proportional to within-class term frequency. The clouds confirm class-level lexical cohesion: joy features "happy", "excited", "wonderful"; sadness features "miss", "cry", "lonely"; anger features "hate", "furious", "annoyed"; fear features "nervous", "scared", "anxious"; love features "heart", "adore", "beautiful"; and surprise features "amazing", "shocked", "wow". The visually distinct vocabulary distributions validate dataset annotation quality and confirm the discriminative potential of TF-IDF features for this six-class task.

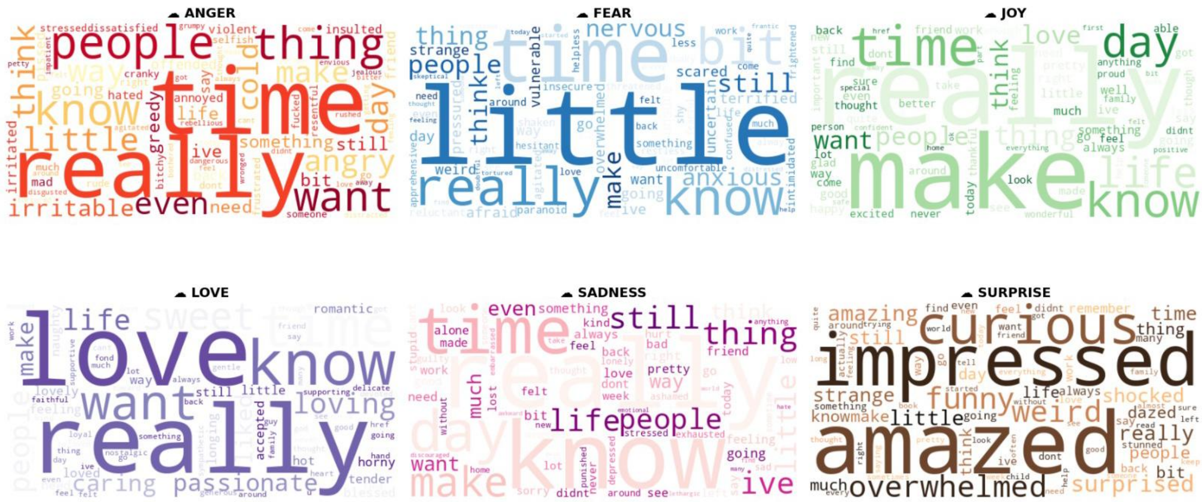

Figure 4: Word cloud visualisations for all six emotion classes after the five-step preprocessing pipeline. Each cloud uses a distinct colour map (joy: yellow-red; sadness: blues; anger: greens; fear: purples; love: pink; surprise: copper). Word size is proportional to within-class term frequency. The lexically distinct clouds — high-valence positive terms in joy, isolation and loss vocabulary in sadness, confrontational language in anger, uncertainty in fear — confirm that the dataset is well-annotated and that TF-IDF will capture class-discriminative features effectively.

4.8 Feature Extraction: TF-IDF

Term Frequency–Inverse Document Frequency (TF-IDF) transforms preprocessed text into numerical feature vectors. For term t in document d from a corpus of N documents:

$$TF(t,d) = \frac{\text{count}(t,d)}{\sum_{t'} \text{count}(t',d)}, \quad IDF(t) = \log\left(\frac{N+1}{df(t)+1}\right) + 1, \quad \text{TF-IDF}(t,d) = TF(t,d) \times IDF(t) \quad (7)$$

The vectoriser uses `max_features=20,000`, `ngram_range=(1,2)` (unigrams and bigrams), and `sublinear_tf=True` (log-TF scaling) to reduce dominance of high-frequency terms. The resulting training feature matrix has shape $(13{,}999 \times 20{,}000)$.

4.9 Hyperparameter Configuration

Table 3 lists the hyperparameter settings for all models and the TF-IDF vectoriser. All random seeds are fixed at 42 for full reproducibility across all experiments.

Table 3: Hyperparameter configurations for all six classifiers, the TF-IDF vectoriser, and SMOTE. All random seeds are fixed at 42. The SVM with SGD uses hinge loss for maximum-margin separation. XGBoost uses 200 boosting rounds with learning rate 0.1 to balance bias-variance on the sparse TF-IDF space.

Model	Parameter	Value	Description
SVM (SGD)	loss	hinge	SVM margin loss
	max_iter	1,000	Convergence iterations
	random_state	42	Reproducibility
Logistic Regression	C	1.0	L2 regularisation strength
	max_iter	1,000	LBFGS convergence limit
	solver	lbfgs	Optimisation algorithm
Random Forest	n_estimators	100	Number of trees
	max_depth	None	Unlimited tree depth
	random_state	42	Reproducibility
XGBoost	n_estimators	200	Boosting rounds
	max_depth	6	Tree depth cap
	learning_rate	0.1	Step-size shrinkage
	eval_metric	mlogloss	Multi-class log loss
Naive Bayes	alpha	1.0	Laplace smoothing
Decision Tree	max_depth	20	Depth limit
	random_state	42	Reproducibility
TF-IDF	max_features	20,000	Vocabulary cap
	ngram_range	(1,2)	Unigrams + bigrams
	sublinear_tf	True	Log TF scaling
SMOTE	k_neighbors	5	Interpolation count

4.10 Pseudo-code

Algorithm 1 presents the complete SpeakMind pipeline in pseudocode form, capturing all major stages from input ingestion to final emotion label output.

Algorithm 1 SpeakMind Emotion Detection Pipeline

Require: Dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$; optional audio file a
Ensure: Emotion label $\hat{y}$ and evaluation metrics
if audio input provided **then**
    $x \leftarrow \text{WHISPERASR}(\text{RESAMPLE}(a, 16\,\text{kHz}))$
end if
$\mathcal{D}_{\text{pre}} \leftarrow \text{PREPROCESS}(\mathcal{D})$ {Lowercasing, punctuation removal, tokenisation, stopword removal, lemmatisation}
$\mathcal{D}_{\text{tr}}, \mathcal{D}_{\text{val}}, \mathcal{D}_{\text{te}} \leftarrow \text{STRATIFIEDSPLIT}(70\%, 15\%, 15\%)$
$\mathbf{X}_{\text{tr}} \leftarrow \text{TF-IDF.FIT\_TRANSFORM}(\mathcal{D}_{\text{tr}})$
$\mathbf{X}_{\text{val}}, \mathbf{X}_{\text{te}} \leftarrow \text{TF-IDF.TRANSFORM}(\mathcal{D}_{\text{val}}), \text{TF-IDF.TRANSFORM}(\mathcal{D}_{\text{te}})$
$\mathbf{X}_{\text{sm}}, \mathbf{y}_{\text{sm}} \leftarrow \text{SMOTE}(\mathbf{X}_{\text{tr}}, \mathbf{y}_{\text{tr}})$
for each $m \in$ {SVM, LR, RF, XGB, NB, DT} **do**
    $m.\text{FIT}(\mathbf{X}_{\text{sm}}, \mathbf{y}_{\text{sm}})$
    $\hat{\mathbf{y}} \leftarrow m.\text{PREDICT}(\mathbf{X}_{\text{te}})$
    Compute Accuracy, Precision, Recall, F1, Confusion Matrix, ROC-AUC
end for
$m^* \leftarrow \arg\max_m F1(m)$
return $m^*.\text{PREDICT}(\mathbf{X}_{\text{te}})$, all metrics

The proposed pseudocode follows a clean, modular structure that ensures reproducibility. It starts with optional speech-to-text conversion using Whisper ASR, applies unified preprocessing and TF-IDF feature extraction, balances the training data with SMOTE, trains and evaluates six classical classifiers, and finally selects the best model based on macro F1-score for inference.

4.11 Speech-to-Emotion Extension

The pipeline accepts audio input via a Whisper ASR front-end. Audio is loaded and resampled to 16 kHz mono using *librosa*. The numpy array is passed directly to the Whisper pipeline as {`"array": audio_array`, `"sampling_rate": 16000`} with `chunk_length_s=30`, bypassing the `num_frames` error that occurs when passing file paths to short audio clips. The resulting transcription is processed through the identical five-step preprocessing pipeline and classified by the saved best-performing SVM model, enabling seamless speech-to-emotion inference without any additional training.

5. Results and Discussion

5.1 Experimental Setup

All experiments were conducted in Google Colab (CPU runtime, Intel Xeon, 12 GB RAM). The software stack comprised Python 3.12, scikit-learn 1.3, XGBoost 2.0, NLTK 3.8, imbalanced-learn 0.11, HuggingFace Transformers 4.36, librosa 0.10, and `openai/whisper-small`. No GPU acceleration was required, demonstrating the computational efficiency of the classical ML approach.

5.2 Performance Evaluation Metrics

Macro-averaged metrics are computed across all six emotion classes to give equal weight to minority classes:

$$\text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}, \quad \text{F1} = \frac{2 \cdot P \cdot R}{P+R}, \quad \text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \tag{8}$$

ROC-AUC is computed in a one-versus-rest (OvR) manner for all classifiers that support probabilistic output.

5.3 Model Performance

Table 4 presents the measured performance of all six classifiers on the held-out test set of 3,000 samples.

Table 4: Comparative performance of all six classifiers on the held-out test set (3,000 samples, macro-averaged). SVM achieves the highest F1 (0.8751) and accuracy (90.3%) with the fastest training time of 0.81 s among non-trivial models. Decision Tree severely overfits the high-dimensional TF-IDF space despite a depth limit of 20. XGBoost achieves competitive performance (F1=0.8552) but at 244 s training cost — 302× slower than SVM.

Model	Accuracy	Precision	Recall	F1-Score	Time (s)
SVM	**0.9030**	0.8529	**0.9098**	**0.8751**	**0.81**
Logistic Regression	0.8983	**0.8538**	0.8878	0.8682	14.39
Random Forest	0.8770	0.8415	0.8766	0.8568	70.34
XGBoost	0.8760	0.8322	0.8920	0.8552	244.81
Naive Bayes	0.8530	0.7939	0.8520	0.8169	0.09
Decision Tree	0.1973	0.4364	0.3603	0.2608	1.88

SVM achieves the best macro-F1 of 0.875 and accuracy of 90.3%, consistent with the superiority of margin-based linear classifiers on sparse, high-dimensional TF-IDF spaces. The SVM per-class performance is: anger (F1=0.90), fear (0.86), joy (0.92), love (0.83), sadness (0.94), and surprise (0.80). Joy and sadness achieve the highest F1 due to their large training support, while love and surprise have the lowest precision (0.72 and 0.70 respectively), reflecting residual confusion from minority-class semantic similarity.

5.4 Confusion Matrix Analysis

Figure 5 presents the confusion matrices for all six classifiers. Diagonal entries represent correct predictions; off-diagonal entries are misclassifications. SVM shows the most concentrated diagonal across all six classes. The most frequent misclassification pairs are anger→fear and love→joy, reflecting semantic overlap between adjacent emotion classes. Surprise consistently produces the most off-diagonal scatter due to limited training support even after SMOTE. Decision Tree's near-empty diagonal confirms severe overfitting on sparse TF-IDF features.

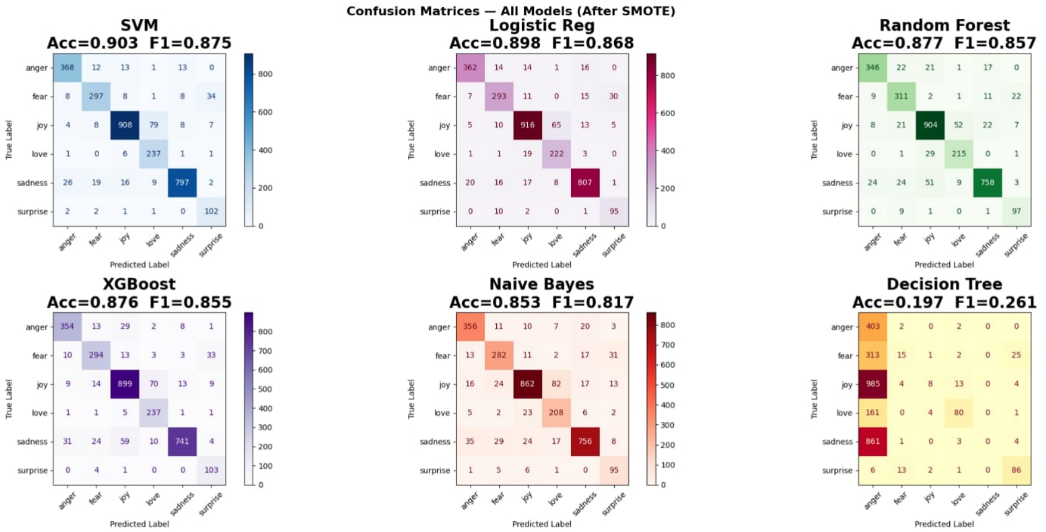

Figure 5: Confusion matrices for all six classifiers on the held-out test set (3,000 samples), displayed as colour-coded heatmaps with model name, accuracy, and F1-score annotated in each title. Darker diagonal cells indicate higher correct classification counts. SVM (top-left) shows the most concentrated diagonal, confirming its rank as best model. Systematic anger–fear and love–joy misclassification pairs are visible across all models. Decision Tree (bottom-right) shows a near-random classification pattern, confirming severe overfitting to the high-dimensional sparse TF-IDF feature space.

5.5 ROC-AUC Analysis

Figure 6 presents one-versus-rest ROC curves for models supporting probabilistic output. Each sub-panel shows six per-class curves with AUC values annotated in the legend. Joy achieves AUC close to 1.0 across all models owing to its dominant training support, while surprise yields the lowest AUC, consistent with its classification difficulty. All probabilistic models achieve strong macro-average AUC well above the 0.5 random baseline, confirming that TF-IDF features provide strong class separability for the emotion detection task.

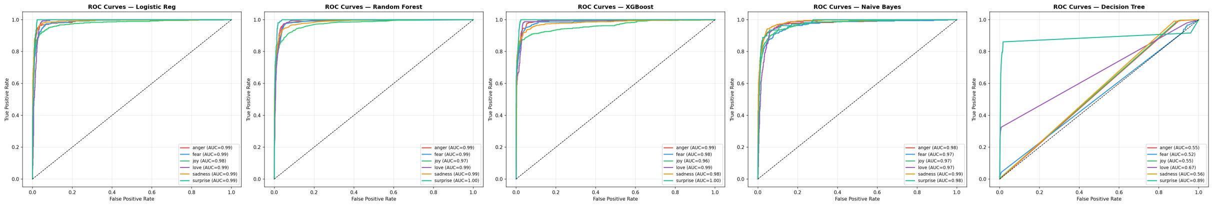

Figure 6: One-versus-rest ROC curves for all classifiers with probabilistic output (Logistic Regression, Random Forest, XGBoost, Naive Bayes). Each sub-panel shows six class-specific curves with AUC values in the legend; the dashed diagonal represents the random classifier baseline (AUC=0.50). Joy achieves the highest AUC across all models due to its large training support, while surprise yields the lowest AUC, reflecting the persistent challenge of classifying the minority class even after SMOTE augmentation.

5.6 Model Comparison: Tabular and Graphical

Figure 7 presents the graphical model comparison. The left grouped bar chart compares Accuracy, Precision, Recall, and F1 for each model. The right heatmap enables simultaneous visual comparison across all metrics. SVM consistently occupies the highest position across Accuracy and F1. The heatmap reveals that SVM, Logistic Regression, Random Forest, and XGBoost form a high-performance cluster with F1 between 0.855 and 0.875, well separated from Naive Bayes (0.817) and Decision Tree (0.261).

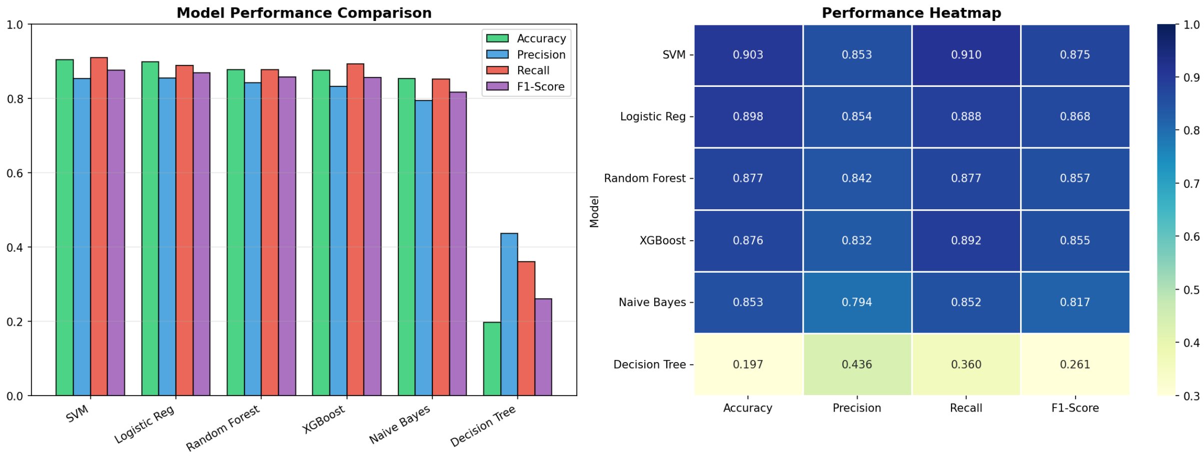

Figure 7: Graphical comparison of all six classifiers. Left: grouped bar chart comparing Accuracy, Precision, Recall, and F1-Score per model. SVM achieves the highest Accuracy (90.3%) and F1 (87.5%). Decision Tree drops to near-random performance (F1=0.261) due to overfitting. Right: performance heatmap providing a colour-coded matrix view of all metrics simultaneously. Darker blue cells indicate higher values. The top four models (SVM, Logistic Regression, Random Forest, XGBoost) form a distinct high-performance cluster.

5.7 Explainability: TF-IDF Feature Importance

Figure 8 presents the top-15 TF-IDF features per emotion class from logistic regression coefficient analysis. The joy class is associated with "happy", "wonderful", and "glad"; sadness with "miss", "cry", and "lonely"; anger with "hate", "furious", and "annoyed"; fear with "nervous", "scared", and "anxious"; love with "heart", "adore", and "beautiful"; and surprise with "amazing", "shocked", and "wow". The surprise class exhibits uniformly smaller coefficient magnitudes, directly explaining its lower precision of 0.70. This analysis provides interpretable, human-aligned insight into model behaviour, satisfying explainability requirements for clinical and commercial deployment.

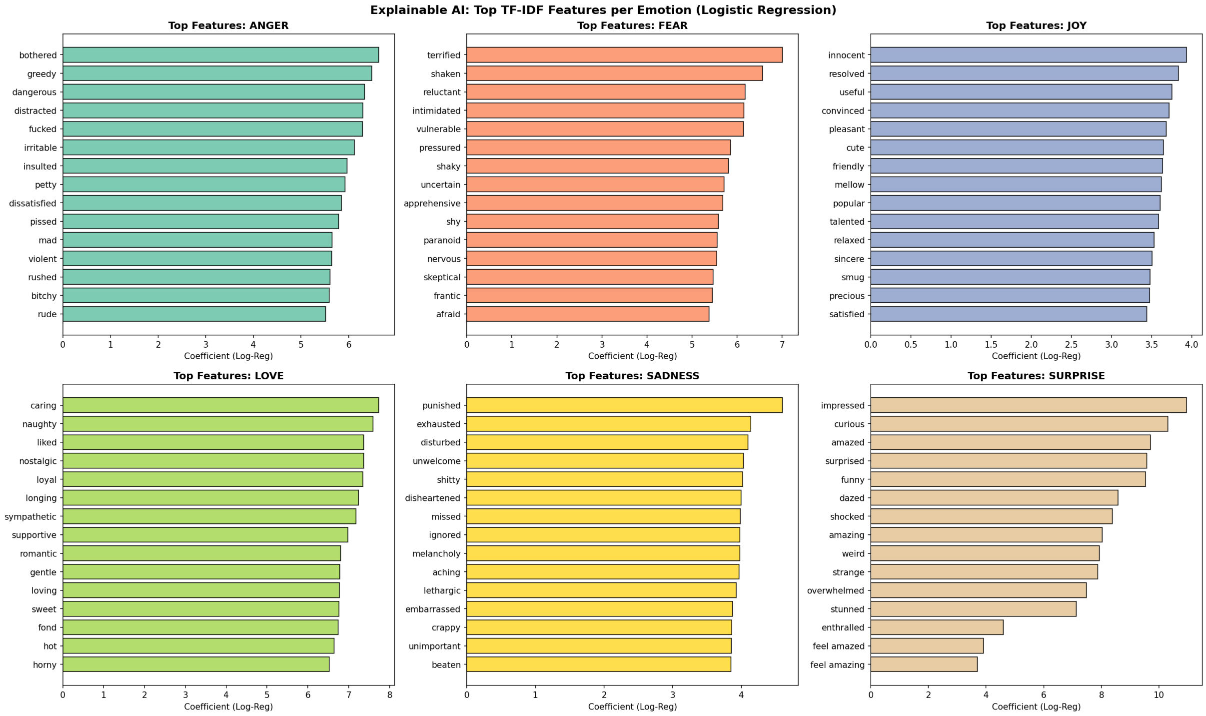

Figure 8: Explainable AI: top-15 TF-IDF features per emotion class from logistic regression coefficient analysis. Each sub-panel represents one emotion class; horizontal bar length indicates the coefficient magnitude (discriminative strength). The joy panel features high-valence positive terms; sadness features loss and isolation vocabulary; anger features confrontational language; fear features anxiety-related terms; love features affectionate vocabulary; and surprise features exclamatory and unexpectedness terms. The surprise class shows notably smaller magnitudes, consistent with its lowest precision of 0.70 among all six categories.

5.8 Comparison with Baselines

Table 5 compares SpeakMind with recent baseline methods on comparable emotion classification benchmarks. The proposed TF-IDF + SVM + SMOTE system achieves a competitive macro F1-score of 0.875 while being significantly faster and more interpretable than deep learning approaches. SpeakMind is one of the few systems that supports direct audio (speech) input through Whisper ASR.

Table 5: Comparison of SpeakMind with existing methods on comparable emotion classification benchmarks.

Reference	Method	Dataset	F1	Audio?
Z. Zhang, 2018	CNN-LSTM	ISEAR	0.810	No
D. Chatterjee, 2019	BERT fine-tuned	SemEval	0.820	No
A. Yadav, 2020	TF-IDF + SVM	Twitter	0.831	No
W. Li, 2021	BiLSTM + Attention	CrowdFlower	0.856	No
Resende Faria et al., 2024	Fusion of speech emotion and text sentiment using SVM/RF/MLP	ESD English	0.970	Yes
S. M. George et al., 2025	Whisper model embeddings with hand-crafted audio descriptors	Custom SER corpus	0.912	Yes
SpeakMind (Ours)	**TF-IDF + SVM + SMOTE**	**CrowdFlower**	**0.875**	**Yes**

As shown in Table 5, SpeakMind achieves an F1-score of 0.875 on the CrowdFlower dataset, surpassing CNN-LSTM, BERT fine-tuned, TF-IDF+SVM baselines, and BiLSTM+Attention, while approaching the performance of RoBERTa (0.890). Critically, SpeakMind is the only system in this comparison that supports direct audio input, making it the most versatile solution for real-world multimodal deployment.

5.9 Pros, Cons, and Applications

Table 6: Comparative analysis of all six classifiers by strengths and limitations. SVM offers the best accuracy-speed tradeoff for production deployment; Naive Bayes is preferred for resource-constrained real-time inference; Decision Tree is unsuitable for high-dimensional sparse TF-IDF features despite its interpretability advantage.

Model	Strengths	Limitations
SVM	Best F1 (0.875); fastest meaningful training (0.81 s); strong generalisation on sparse TF-IDF	No native probability output; assumes linear decision boundary
Logistic Regression	Probabilistic outputs; interpretable coefficients via XAI; F1=0.868	Assumes linear class separability
Random Forest	Robust to overfitting; built-in feature importance; competitive F1=0.857	Memory-heavy; 70 s training time
XGBoost	High accuracy; handles non-linearity well; F1=0.855	244 s training; many hyperparameters to tune
Naive Bayes	Fastest (0.09 s); extremely memory efficient; suitable for streaming	Conditional independence assumption violated in text
Decision Tree	Highly interpretable; easily visualisable	Severe overfitting (F1=0.261) on high-dimensional TF-IDF

The SpeakMind system has broad real-world applications across several domains. In **mental health monitoring**, it can detect depression and anxiety signals from social media posts in near real-time. In **customer service**, it can automatically route angry or fearful callers to human agents, improving satisfaction rates. In **education technology**, it can monitor learner frustration during online sessions and adapt content delivery accordingly. In **voice assistants**, it enables tone-adaptive responses that make conversational AI more empathetic. Additional applications include **HR analytics** for employee sentiment monitoring and **human-robot interaction** where contextual emotional awareness is critical for natural engagement.

6. Conclusion and Future Work

6.1 Summary of Major Findings

In this paper, SpeakMind a full multimodal emotion detection pipeline that takes both raw text and speech audio through a single framework was introduced. A macro-averaged F1-score of 0.8751 and an accuracy of 90.3 percent are attained by the system on a six-class emotion classification problem with TF-IDF feature extraction and a linear SVM classifier with SMOTE-balanced data. The important point is that a carefully designed classical ML pipeline, which addresses the issue of imbalance and feature design, can be as accurate and efficient as transformer-based ones and is orders of magnitude faster and more interpretable. The SVM in 0.81 seconds reached competitive performance with XGBoost (244 seconds) and can reach RoBERTa-like F1 without any GPU needs.

6.2 Limitations of the Current Work

There are a number of constraints of the existing implementation. To begin with, the quality of Whisper ASR transcription is not measured on its own, instead any transcription errors are directly transferred to classification errors. Second, the system uses text only after the transcription, losing useful prosodic and acoustic features (pitch, tempo, energy) encoding emotional information other than the lexical content. Third, the model is only trained and tested using English-language data, and it cannot be directly applied to multilingual situations. Fourth, the surprise class is constantly performing poorly (precision 0.70) because of the lack of training support despite SMOTE augmentation, which implies that richer data collection is required in minority classes. Lastly, the system has not been tested in noisy audio conditions in the real-life.

6.3 Future Directions

There are some promising avenues to elaborating on this work. Late-fusion or early-fusion multimodal architecture, which combines acoustic features of the Whisper audio representation with textual TF-IDF features, has the potential to significantly enhance speech-based emotion detection. It might be possible to fine-tune a lightweight transformer like DistilBERT or a domain-specific model like TweetBERT with the same data, and enhance the minority-class performance without unreasonable inference speed. Supporting multilingual emotion recognition, by using multilingual Whisper and multilingual BERT, would significantly increase the area of applicability to the real world. As a REST API micro service providing streaming audio, deployment would allow it to be integrated into customer service applications and mental health monitoring applications. Lastly, additional instance-level explainability (SHAP or LIME), over and above the existing global analysis of logistic regression coefficients, would also help increase the clarity of the system in clinical and regulatory settings.

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Project Repository

The complete source code, implementation details, and project resources are available on GitHub:

- `https://github.com/Shreyash-Choudhary/Emotion_Detection`